Who Develops Diabetes?

A statistical analysis of the Pima Indian Diabetes Database

768
women of Pima heritage, aged 21+

8
predictors measured at examination

34.9%
developed diabetes

Numerical Analysis, End of Module | University of Essex Online | October 2026

A growing problem, measured differently

589M
adults aged 20–79 with diabetes in 2024

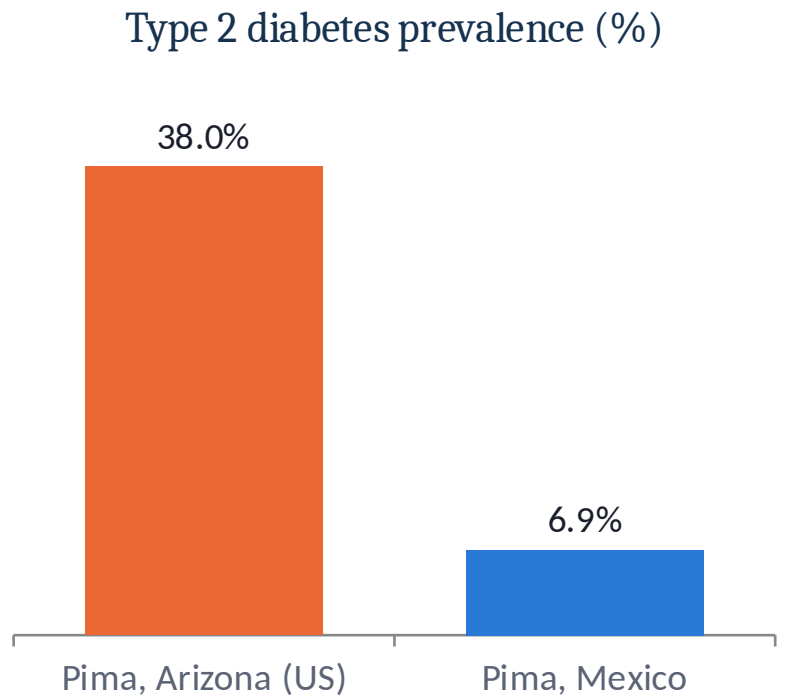
853M
projected by 2050

43%
of adults with diabetes undiagnosed

! NOT STRAIGHTFORWARD

Same disease, two counts

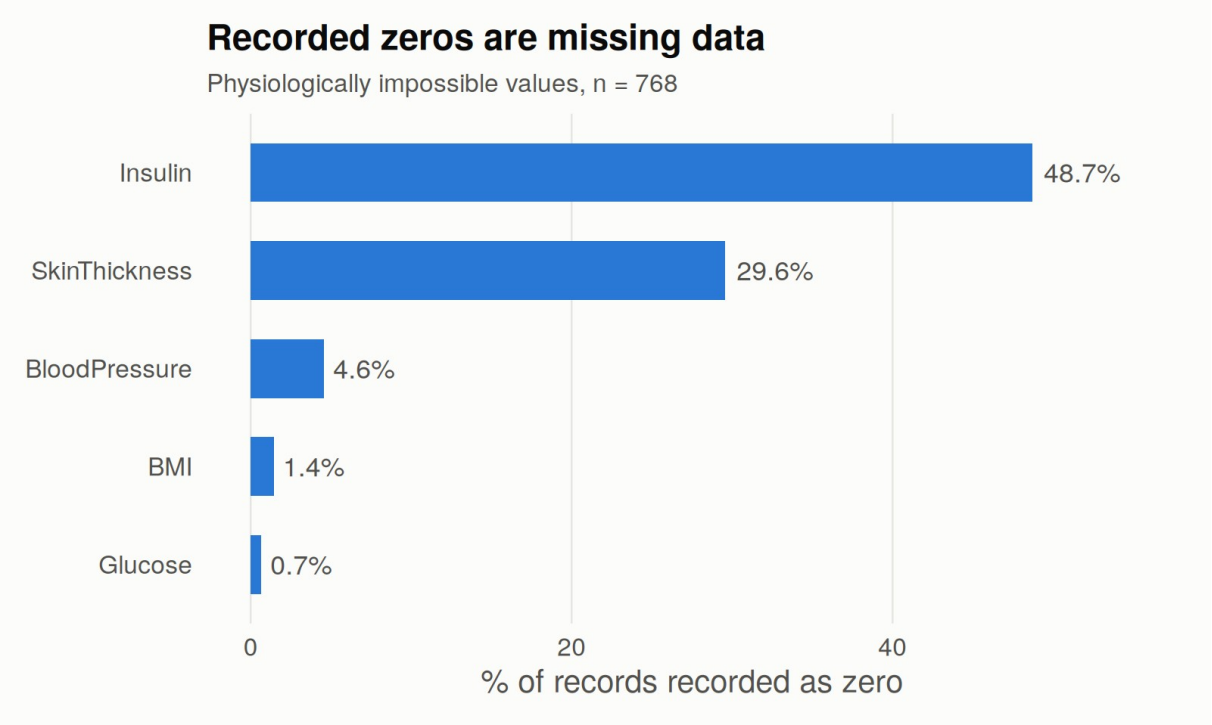
- International Diabetes Federation (2025): 589 million adults aged 20–79
- NCD Risk Factor Collaboration (2024): 828 million adults aged 18+, defined by fasting glucose, HbA1c or medication
- The 239 million gap comes from age range and diagnostic definition, not from the disease itself



Genetically related groups in different environments: lifestyle matters as much as heritage.

Sources: International Diabetes Federation (2025); NCD Risk Factor Collaboration (2024); Schulz *et al.* (2006).

Two data decisions shape every result



Missing values are not related to the outcome, which supports complete-case analysis.

Decision 1: impossible zeros become missing

- Leaving zeros in lowers mean blood pressure from 72.4 to 69.1 mm Hg and raises its SD from 12.4 to 19.4
- Pregnancies = 0 is valid and was kept

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Decision 2: the outcome is defined by glucose

- Diabetes = 2-hour glucose ≥ 200 mg/dL at a follow-up examination (Smith *et al.*, 1988)
- Highest glucose in the data: 199 mg/dL
- Glucose is linked to the outcome by design

Source: Khare (no date), Kaggle; Smith *et al.* (1988). Missingness test: chi-square, Insulin p = 0.29, SkinThickness p = 0.17.

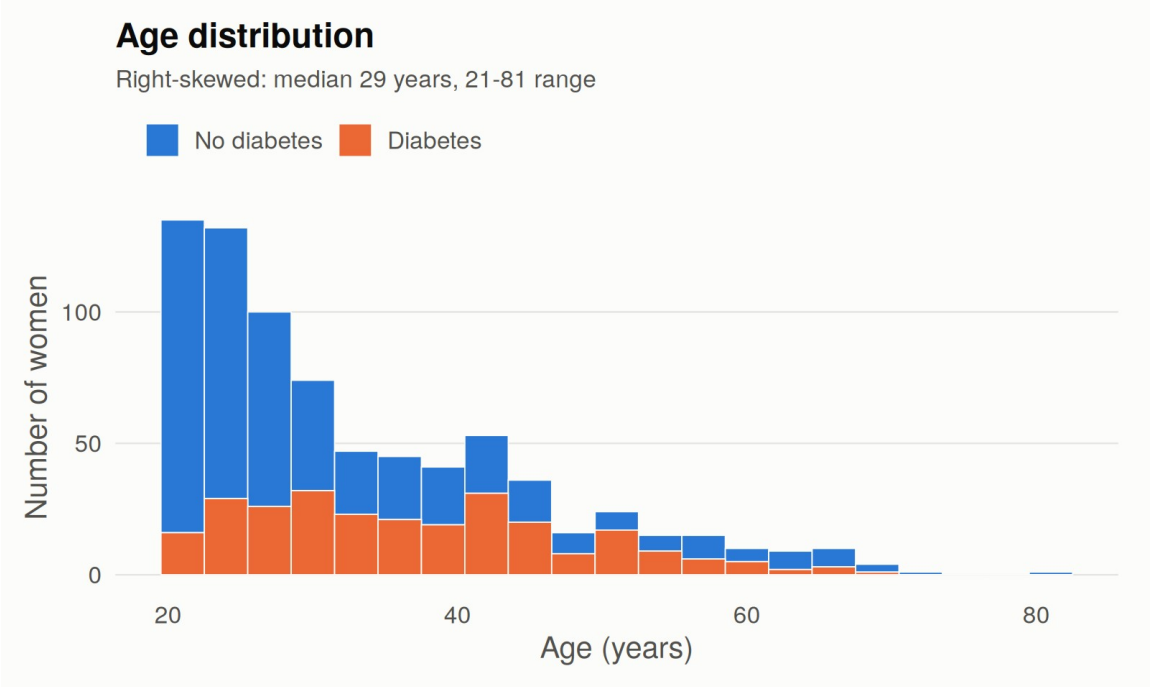
Sample profile: 768 women

768
women in the sample

34.9%
diagnosed with diabetes (268)

29
median age in years (range 21–81)

111 / 657
never pregnant vs pregnant at least once



Right-skewed: skewness 1.13, mean 33.2 > median 29.

Age group	<30	30–39	40–49	50+
Women	396	165	118	89
Share	51.6%	21.5%	15.4%	11.6%

Source: own analysis in R (R Core Team, 2024).

Descriptive statistics

Variable	n	Mean	Median	Mode	Min	Max	Range	SD
Age (years)	768	33.24	29	22	21	81	60	11.76
BMI (kg/m²)	757	32.46	32.3	32.0	18.2	67.1	48.9	6.92
Glucose (mg/dL)	763	121.69	117	99 / 100	44	199	155	30.54
Diastolic BP (mm Hg)	733	72.41	72	70	24	122	98	12.38
Pregnancies (≥ 1)	657	4.49	4	1	1	17	16	3.22

Right skew

Mean exceeds median for age (33.2 vs 29) and pregnancies (4.49 vs 4).

Near symmetry

Blood pressure: mean 72.4, median 72, skewness 0.13.

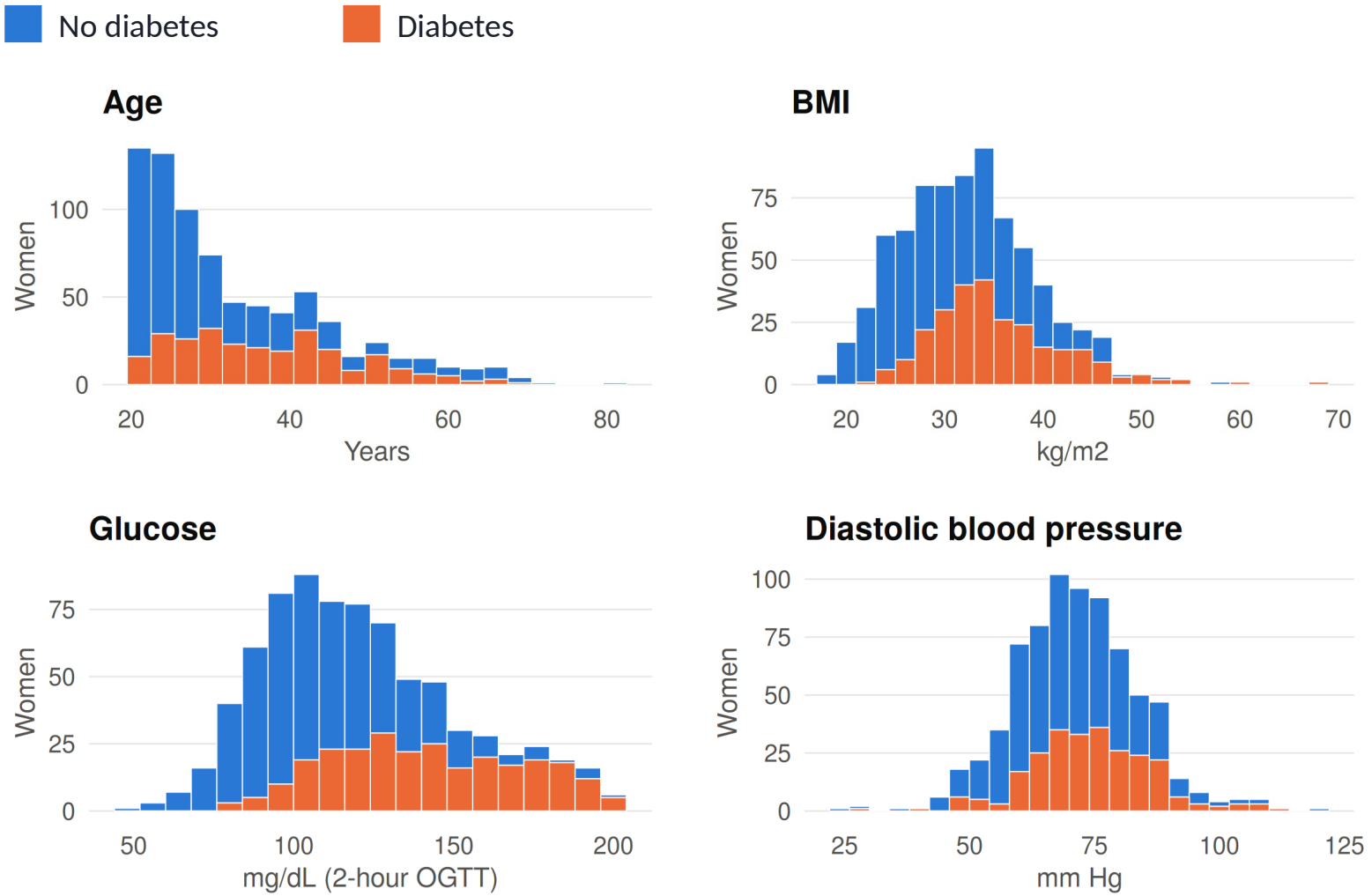
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The mode is fragile

Glucose has two modes (99 and 100); for continuous data the mode depends on rounding.

Zeros treated as missing. Pregnancies summarised for women pregnant at least once (all 768: mean 3.85, median 3, SD 3.37). Dong (2023).

Distributions by diagnosis

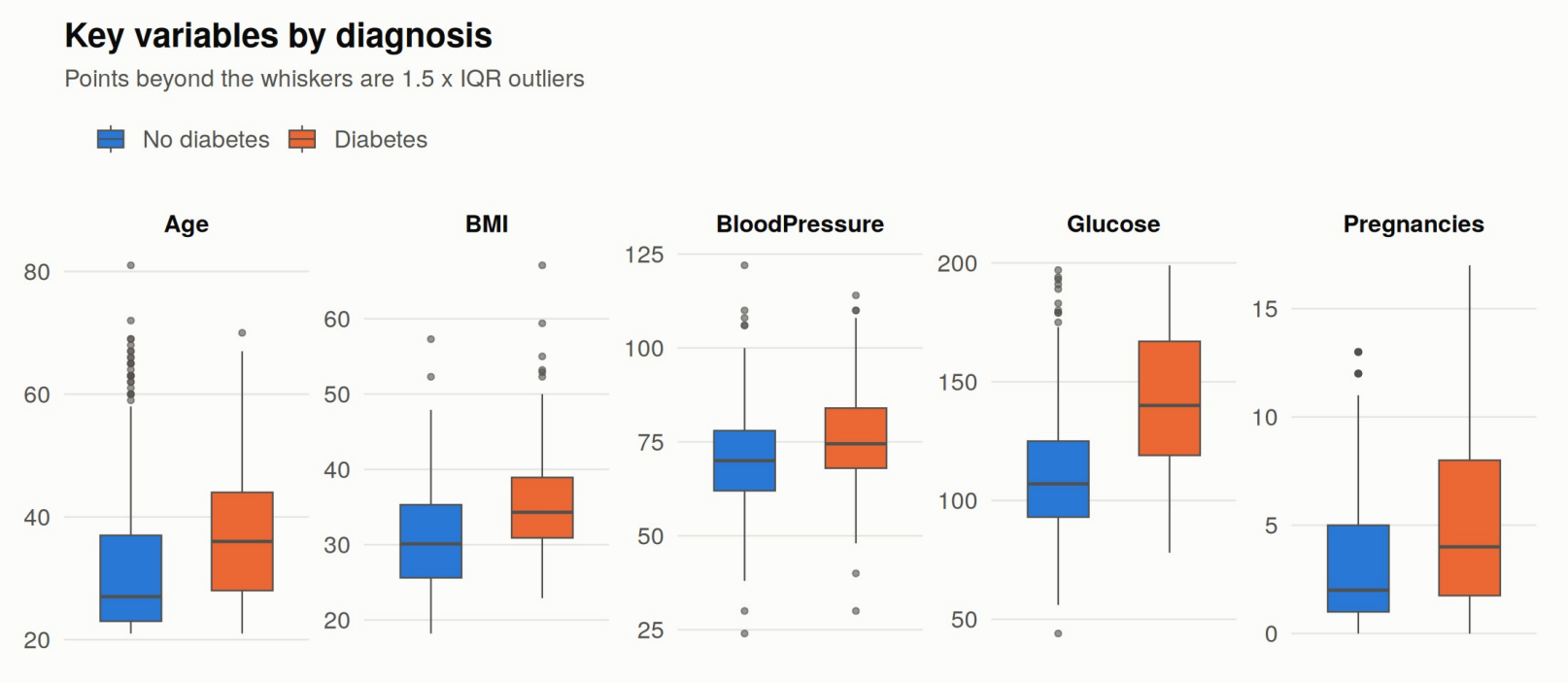


What the charts show

- Glucose separates the groups most clearly
- Blood pressure overlaps heavily
- Bar chart for the discrete pregnancy count

Charts produced in R with ggplot2 (Wickham, 2016); stacked by diagnosis. Bader and Leuzinger (2025).

Outliers: retained, flagged or removed



Variable	High outliers	Maximum
Insulin	24	846 µU/ml
Pedigree function	29	2.42
Age	9	81 years

Rule: values beyond Q1 − 1.5 × IQR or Q3 + 1.5 × IQR. Glucose has no outliers once zeros are removed.

Retained

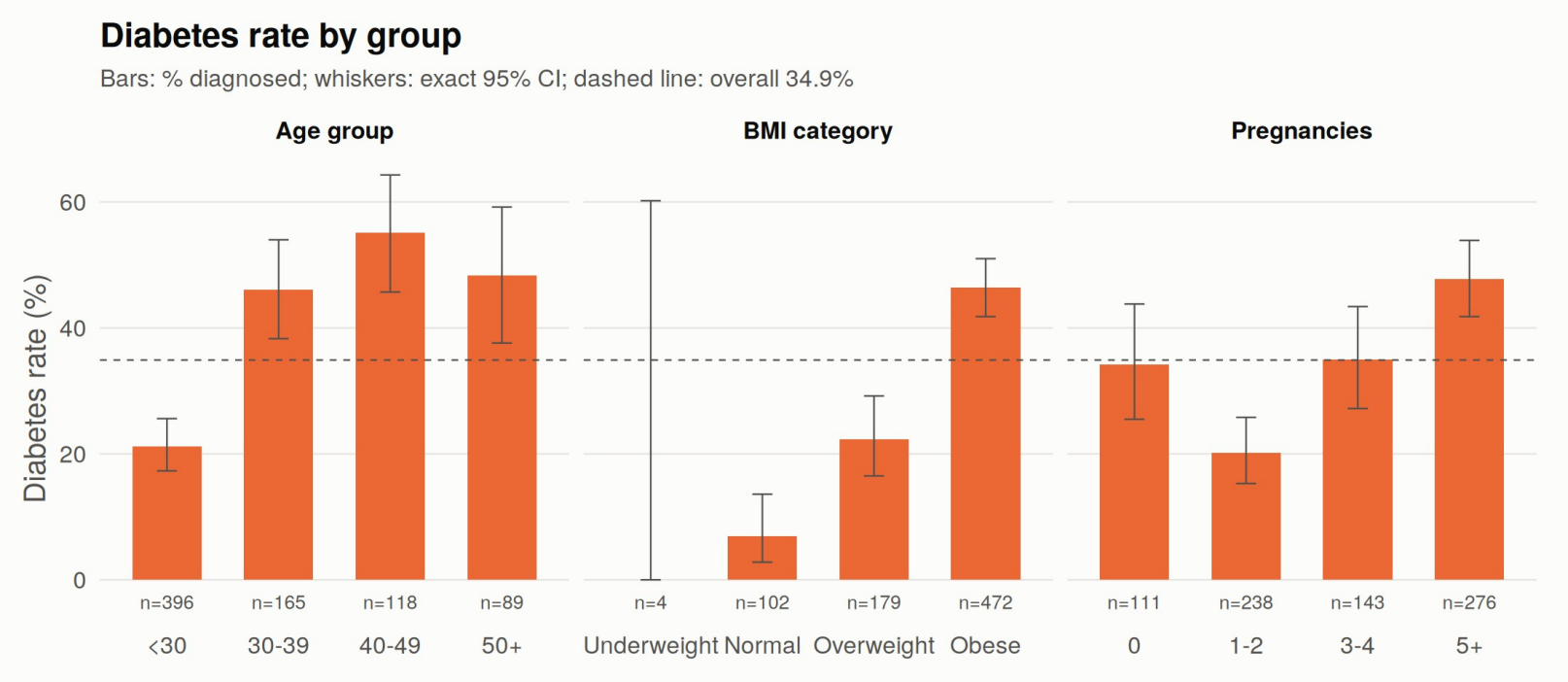
- Extreme but plausible values
- Deleting them would remove the highest-risk patients

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Flagged as probable errors

- Skin-fold of 99 mm (next highest: 63)
- Diastolic BP of 24 and 30 mm Hg

Diabetes rates by group



Underweight: n = 4, 95% CI 0–60%, so no reliable estimate is possible.

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Risk falls after 50

55% at 40–49, then 48% at 50+.

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Pregnancy effect is mostly age

- 5+ pregnancies: OR 1.76, then 1.15 (p = 0.59) after age adjustment
- Mean age 41.9 (5+) vs 27.6 (none)

Exact (Clopper-Pearson) 95% confidence intervals. BMI categories: World Health Organization (2000). 11 women with missing BMI excluded.

Glucose and pregnancies by diagnosis

Glucose

d = 1.19

Cohen's d: very large effect

- Welch t(466) = −14.88, p < 0.001
- Means: 142.3 vs 110.6 mg/dL
- Difference 31.7 (95% CI 27.5 to 35.9)
- Mann-Whitney check: p < 0.001

Pregnancies

r = 0.20

Effect size r = |Z| / √N: small

- Mann-Whitney W = 50,985, p < 0.001
- Medians: 4 vs 2
- Welch t check: p < 0.001, d = 0.48

Test choice

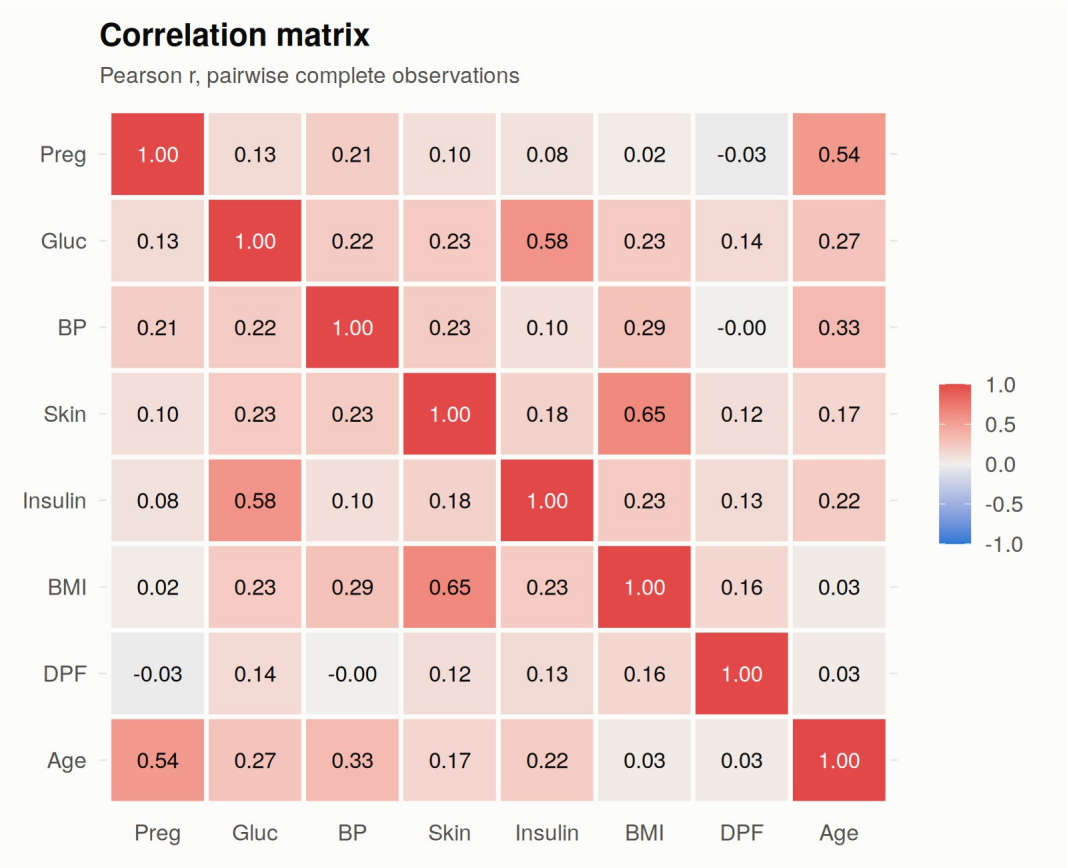
Non-normal data and unequal variances led to Welch and rank-based tests.

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Both results have p < 0.001, but the effect sizes differ greatly.

Assumption checks: Shapiro-Wilk p < 0.001 in both groups; Levene p < 0.001. Wasserstein and Lazar (2016).

Correlation matrix



0.65
BMI and skin thickness

0.58
Glucose and insulin

0.54
Age and pregnancies

Mostly weak

All other pairs are 0.33 or lower.

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Age and pregnancies move together, so regression models struggle to separate them.

Pearson r, pairwise complete observations. Spearman coefficients give the same pattern.

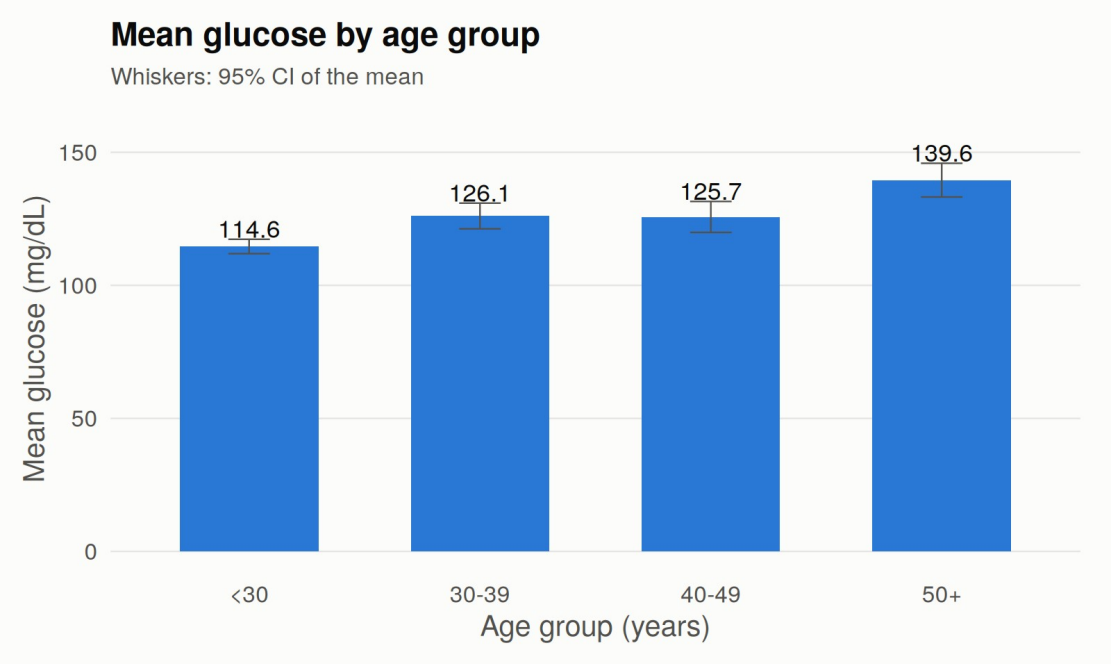
Associations with age and BMI

Test	χ^2 (df)	p	Cramér's V
Outcome × age group	69.9 (3)	< 0.001	0.30
Outcome × BMI category	77.1 (3)	< 0.001	0.32
BMI, merged to 3 groups	77.0 (2)	< 0.001	–
BMI, Fisher exact	–	< 0.001	–

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A small expected count

- Expected count in the underweight cell: 1.4
- Fisher exact and merged tests were added as checks
- All tests give the same conclusion

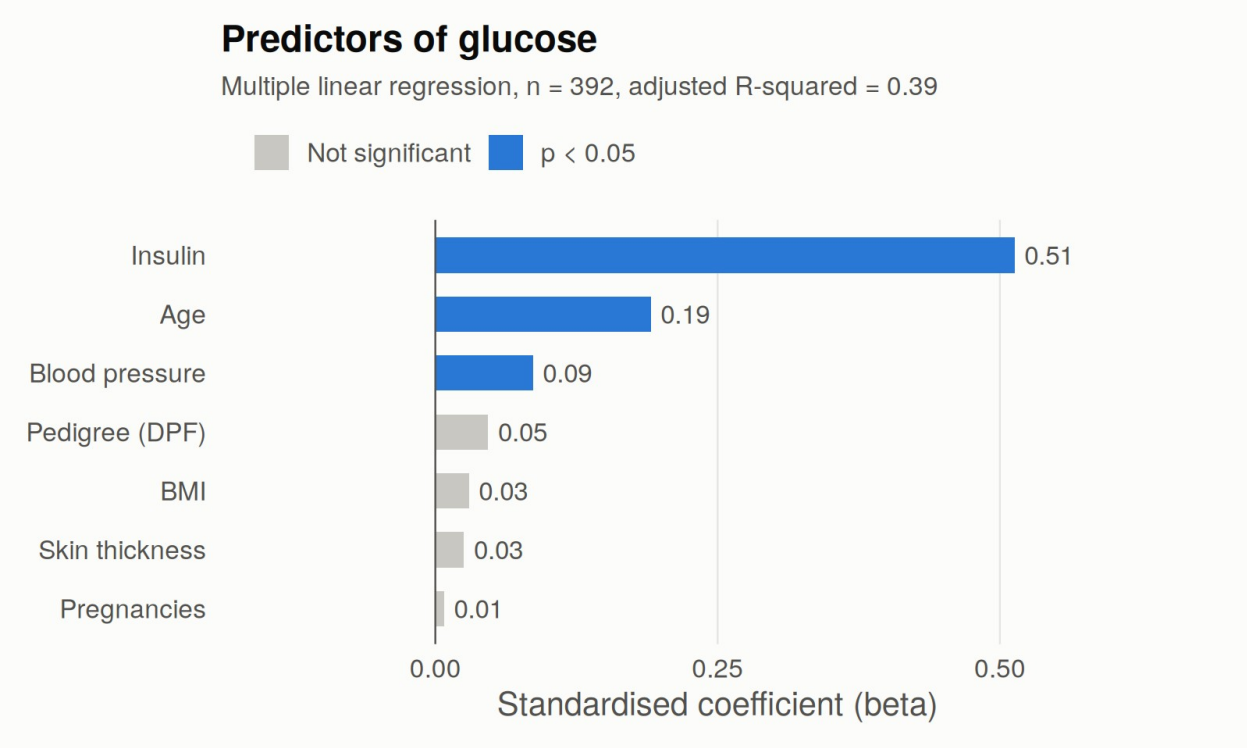


ANOVA: F(3, 759) = 20.44, p < 0.001

η^2 = 0.075. Tukey: every pair differs except 30–39 vs 40–49.

Glucose by age group: Welch F = 20.25 and Kruskal-Wallis p < 0.001 agree with the ANOVA.

Predicting glucose: multiple regression



Significant: insulin (b = 0.133), age (b = 0.58), blood pressure (b = 0.21, p = 0.048).

39%

adjusted R^2, with insulin (n = 392)

13%

without insulin and skin thickness (n = 724)

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Reverse causality

- Glucose triggers insulin release, so the model partly predicts glucose from its consequence
- Insulin alone explains 34% of variance
- Residuals are non-normal and heteroscedastic

n = 392 complete cases; F(7, 384) = 36.8, p < 0.001; all VIF ≤ 2.1. Shapiro-Wilk p < 0.001; Breusch-Pagan p = 0.002.

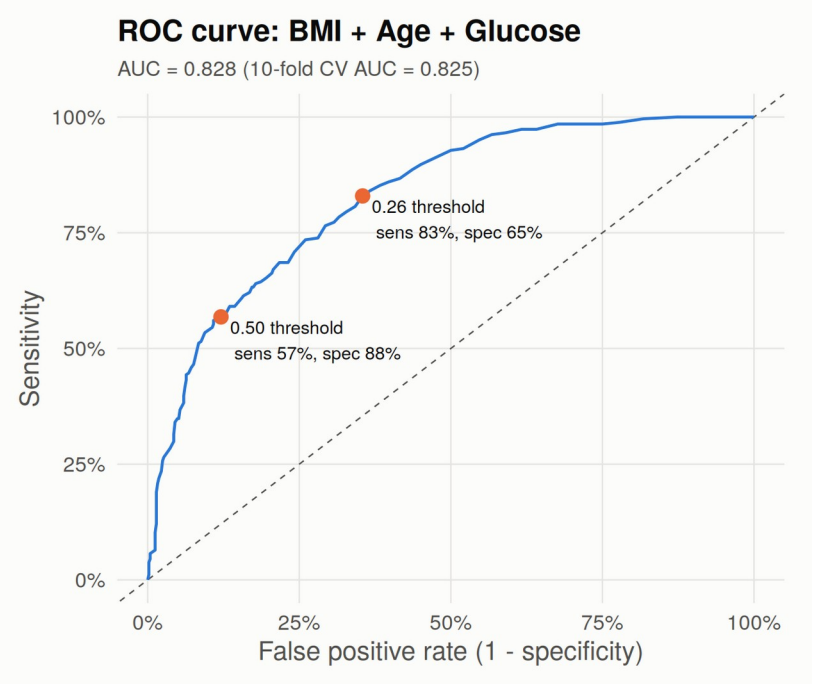
Logistic regression: BMI, age and glucose

1.57
odds ratio per +5 BMI units

1.43
odds ratio per +10 mg/dL glucose

1.33
odds ratio per +10 years of age

Threshold	Accuracy	Sensitivity	Specificity	Missed cases
0.50	77.0%	56.8%	87.9%	114 / 264
0.26	71.0%	**83.0%**	64.5%	45 / 264
0.26 (CV)	71.0%	**82.6%**	64.8%	46 / 264



AUC 0.828; cross-validated AUC 0.825.

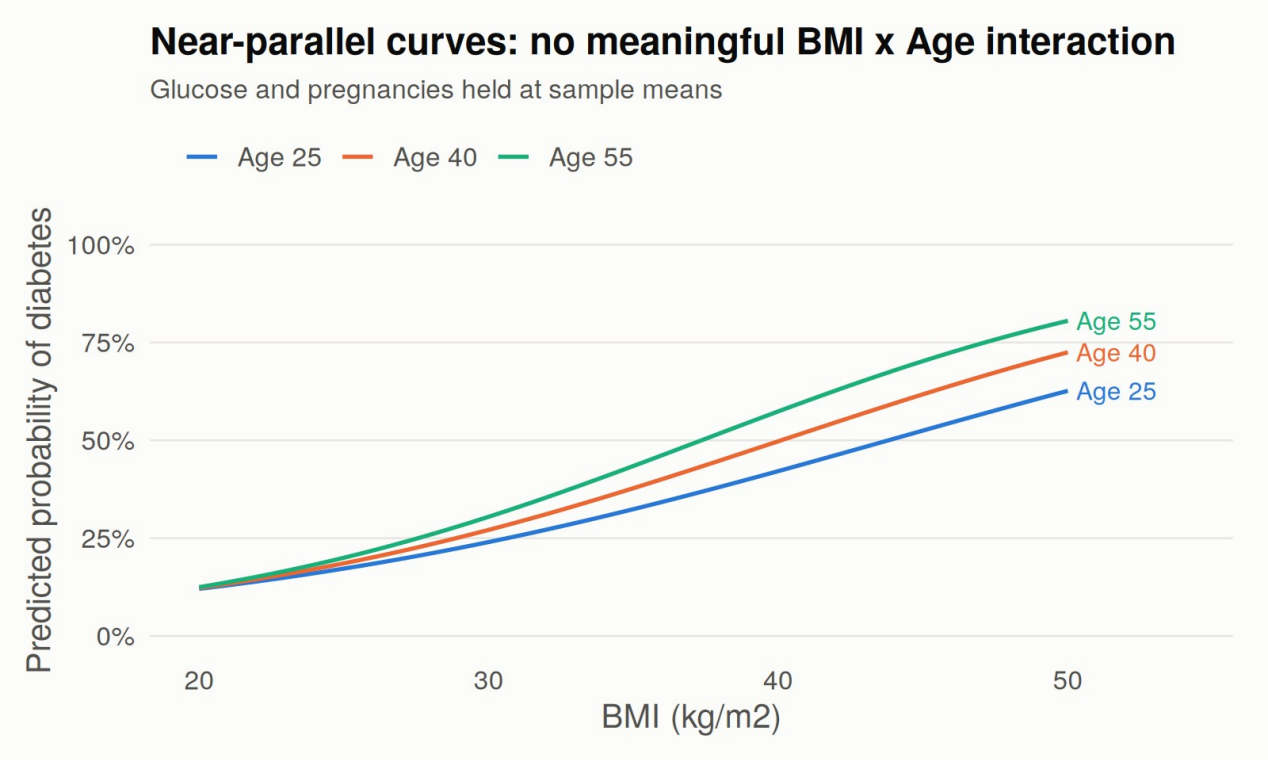
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Goodness of fit depends on grouping

Hosmer-Lemeshow p = 0.197 with 10 groups, but p = 0.047 with 8.

n = 752; all predictors p < 0.001. Nagelkerke R^2 = 0.39. Hosmer *et al.* (1997). 10-fold cross-validation.

BMI × age interaction



Comparison	LR $\chi^2(1)$	p
BMI + Age	0.29	0.59
BMI + Age + Glucose	1.77	0.18
Model 1 vs Model 2	0.56	0.45

Age	OR, +5 BMI	95% CI
25	1.52	1.28–1.81
35	1.59	1.38–1.84
45	1.67	1.36–2.06
55	1.75	1.28–2.40

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Interaction odds ratio 1.001 (95% CI 0.998–1.004) is not significant, so it is not interpreted as an effect.

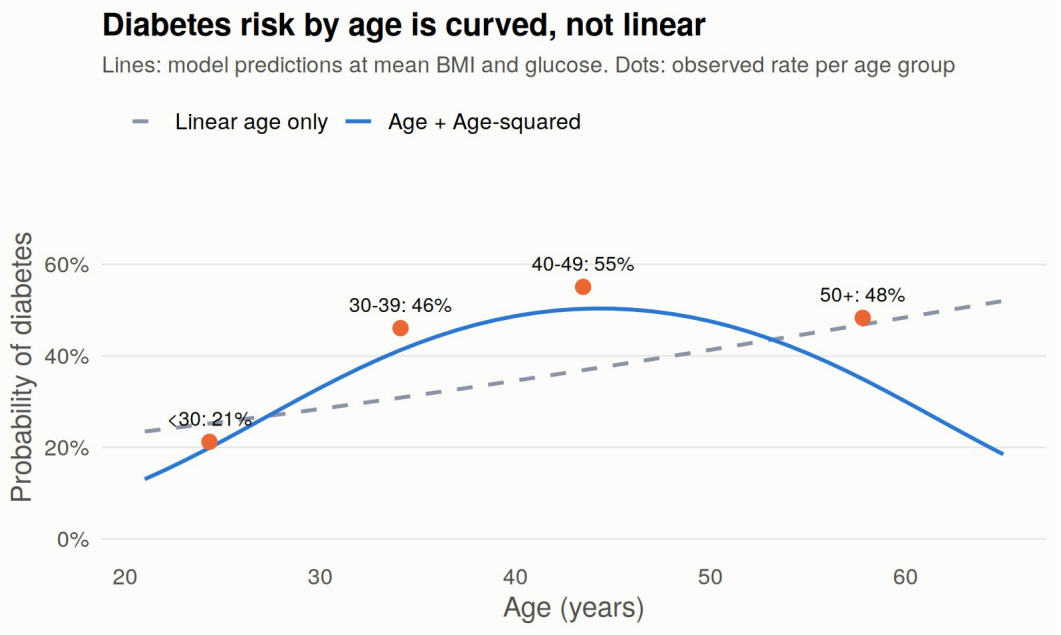
Likelihood ratio tests on n = 752. Predictors centred at mean BMI 32.45 and mean age 33.31.

Model 1 versus Model 2

Metric	Model 1	Model 2
Parameters	5	6
Deviance	712.90	712.33
AIC	**722.90**	724.33
BIC	**746.01**	752.07
AUC	0.834	0.836
Hosmer-Lemeshow p	0.249	0.365
Accuracy	77.0%	77.4%
Sensitivity	57.6%	57.6%
Specificity	87.5%	88.1%

Model 1 preferred

Pregnancies OR 1.12 ($p < 0.001$); age $p = 0.19$, because of collinearity.



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Age acts on a curve

Adding age^2: LR $\chi^2(1) = 33.0$, $p < 0.001$; peak risk near age 44.

Model 1: BMI + Age + Glucose + Pregnancies. Model 2: Model 1 + BMI × Age. Same 752 women.

Linking results to current guidance

47%

of diabetic women are under 35 (126 of 268)

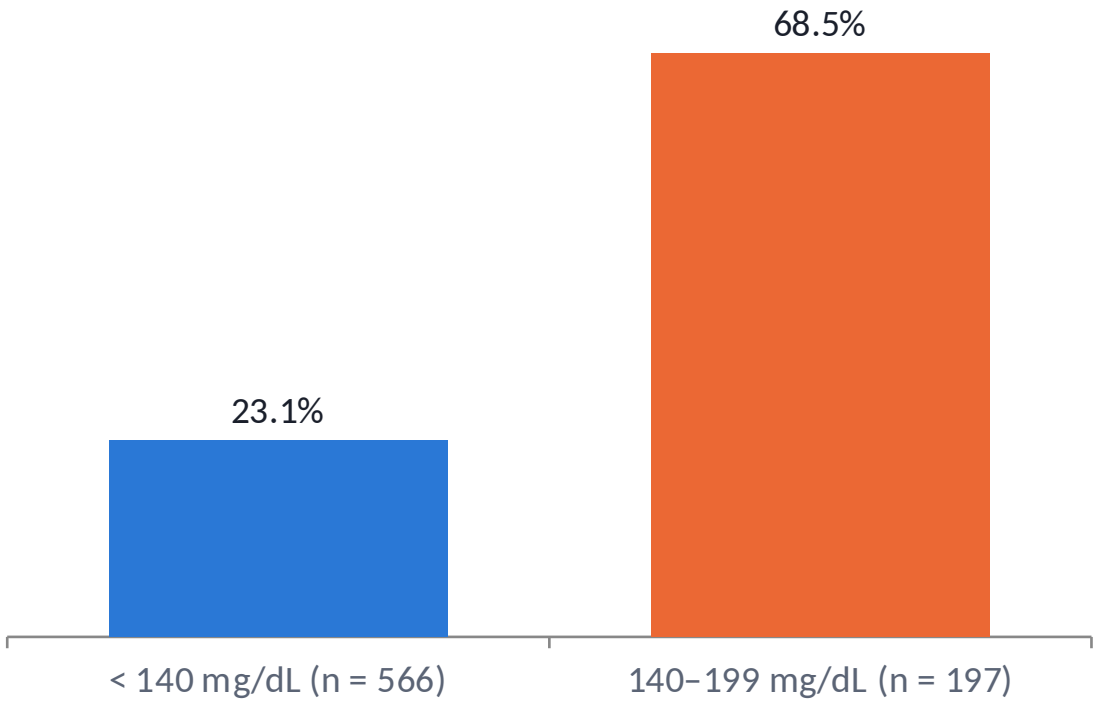
Guidance: test all adults from 35, or earlier with BMI ≥ 25 and a risk factor.

58%

lower incidence with lifestyle change (Diabetes Prevention Program)

Age alone misses many cases in this high-risk group.

Diabetes rate by 2-hour glucose band (%)



140–199 mg/dL is the ADA range for impaired glucose tolerance.

American Diabetes Association Professional Practice Committee (2026); Knowler *et al.* (2002). Glucose bands follow ADA 2-hour OGTT cut-offs.

Limitations and ethics

Generalisability

One population of women, collected decades ago; incidence has shifted over time.

Association, not causation

Insulin results are affected by reverse causality, and half of insulin values are missing.

Indigenous data governance

CARE principles: collective benefit and Indigenous authority over community data.

Reporting and validation

Report models with TRIPOD+AI and validate them externally before clinical use.

Pavkov *et al.* (2007); Carroll *et al.* (2020); Collins *et al.* (2024).

Conclusions and recommendations

Key findings

- Glucose is the strongest marker (d = 1.19; OR 1.43 per 10 mg/dL)
- BMI is the key modifiable factor (OR 1.57 per 5 units)
- Age acts on a curve, peaking near 44
- BMI and age combine additively, not interactively
- Insulin-based results need caution

Recommendations

1. Screen high-risk women from their early thirties, using glucose and BMI
2. Set model thresholds to favour sensitivity
3. Record missing values as missing, and validate models externally

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